A Data-Driven Operational Analysis of Deliveroo's Online Food Delivery Service Using a Publicly Available Delivery Dataset

Dataset:https://www.kaggle.com/datasets/denkuznetz/food-delivery-time-prediction

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D1 – Company Background

1. Organization Overview

Deliveroo is a trailblazing food delivery organization in the United Kingdom that utilizes technology and is based in London but covers large cities such as Birmingham, Manchester, Liverpool, Bristol, and Glasgow. Deliveroo was founded in 2013, and it offers an online service that connects customers to restaurants and grocery partners via web-based and mobile ordering services. The organization works in cooperation with thousands of restaurants and independent delivery riders to provide the delivery of the fast and convenient services of food to homes and workplaces. The objective of Deliveroo is to bring quality meals closer to the user and at the same time operate an efficient and reliable delivery system in the UK. Its service ecosystem involves customer ordering, restaurant cooking, and riders making deliveries during the last miles.

Figure 1Deliveroo UK

Source: https://deliveroo.co.uk/

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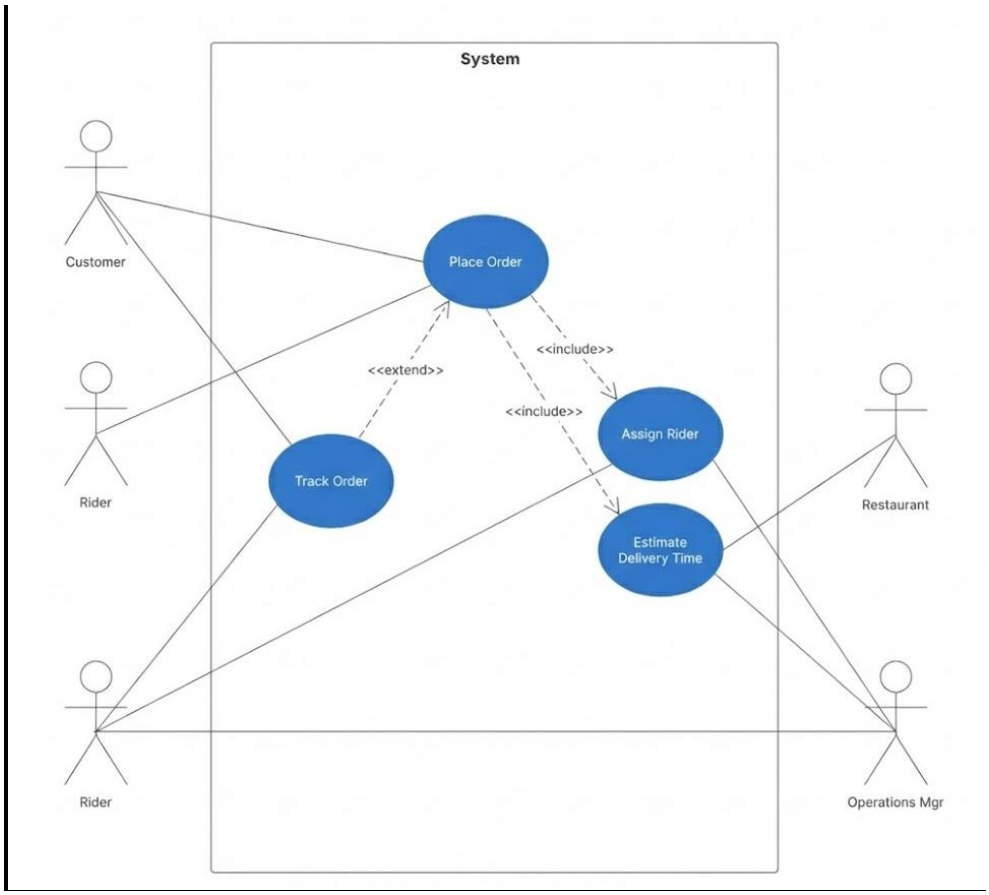

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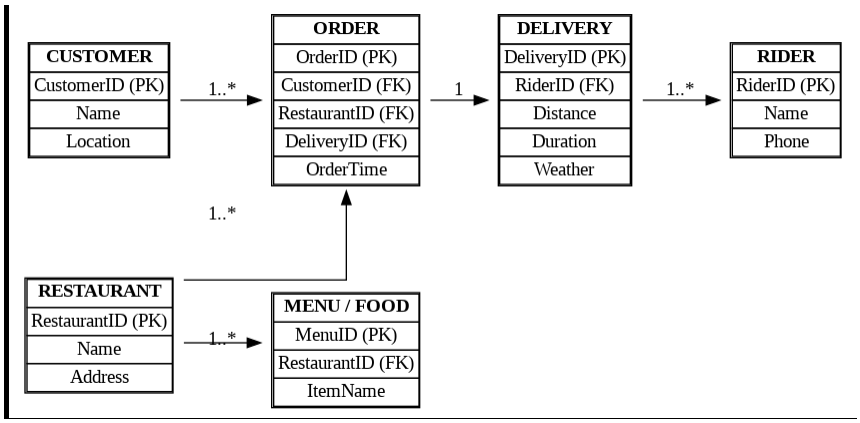

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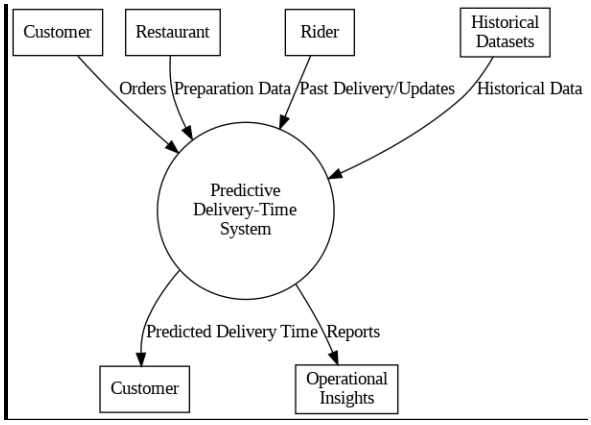

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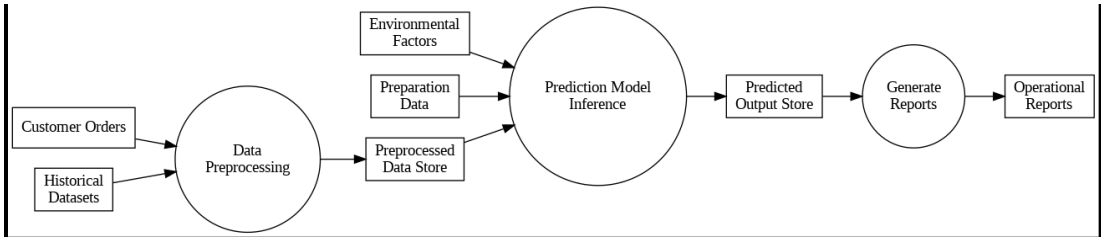

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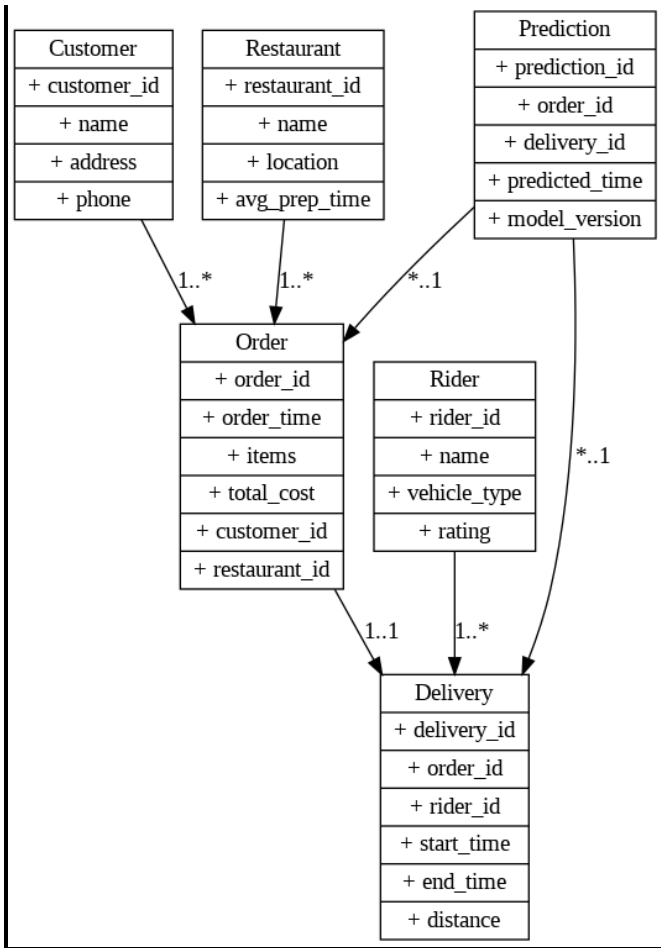

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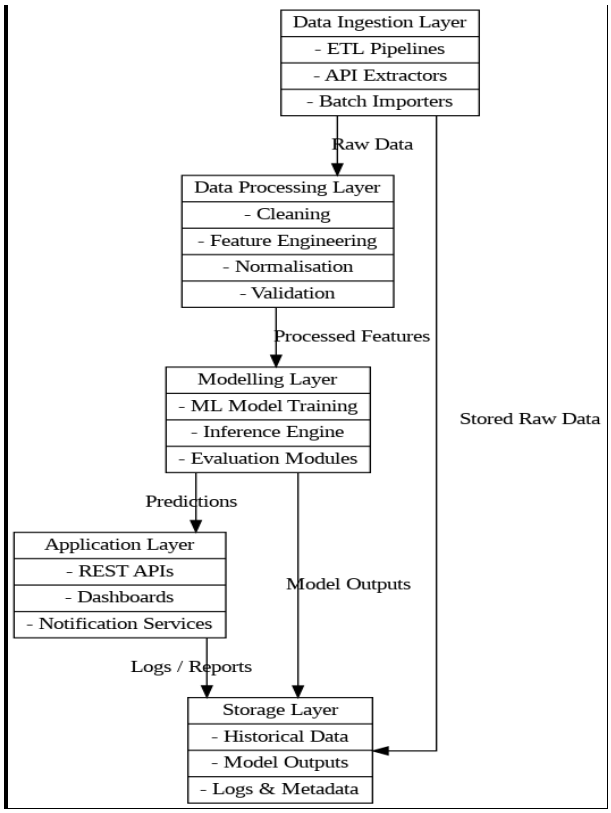

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```
   Order_ID  Distance_km Weather Traffic_Level Time_of_Day Vehicle_Type  \
0       522         7.93   Windy           Low   Afternoon      Scooter
1       738        16.42   Clear        Medium     Evening         Bike
2       741         9.52   Foggy           Low       Night      Scooter
3       661         7.44   Rainy        Medium   Afternoon      Scooter
4       412        19.03   Clear           Low     Morning         Bike

   Preparation_Time_min  Courier_Experience_yrs  Delivery_Time_min
0                    12                     1.0                 43
1                    20                     2.0                 84
2                    28                     1.0                 59
3                     5                     1.0                 37
4                    16                     5.0                 68
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Order_ID                1000 non-null   int64
 1   Distance_km             1000 non-null   float64
 2   Weather                 970 non-null    object
 3   Traffic_Level           970 non-null    object
 4   Time_of_Day             970 non-null    object
 5   Vehicle_Type            1000 non-null   object
 6   Preparation_Time_min    1000 non-null   int64
 7   Courier_Experience_yrs  970 non-null    float64
 8   Delivery_Time_min       1000 non-null   int64
dtypes: float64(2), int64(3), object(4)
memory usage: 70.4+ KB
None

Missing values after cleaning:
 Distance_km               0
Weather                   0
Traffic_Level             0
```

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```
F1 SCORES:
Random Forest: 0.9037656903765691
XGBoost: 0.9159663865546218
Logistic Regression: 0.9029535864978903
```

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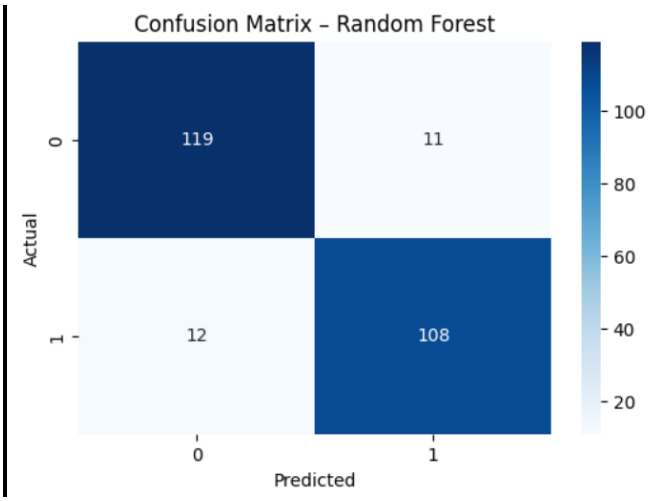

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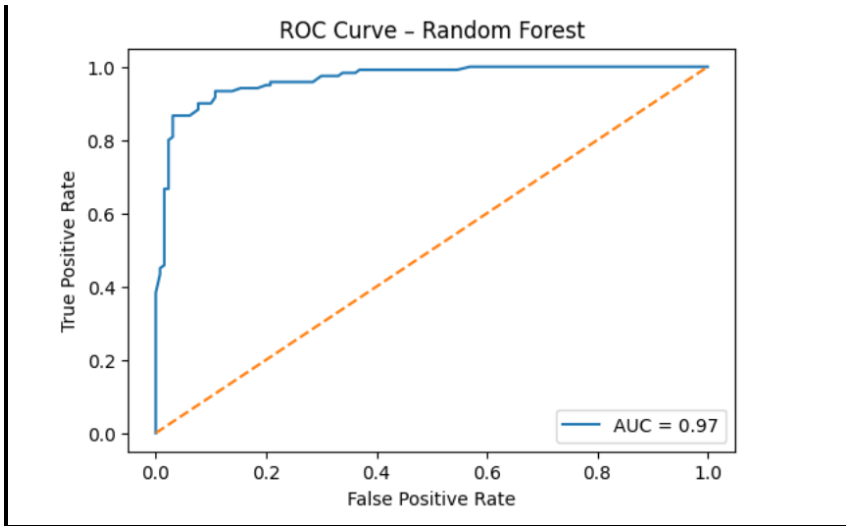
ROC Curve – Random Forest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
AUC = 0.97

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